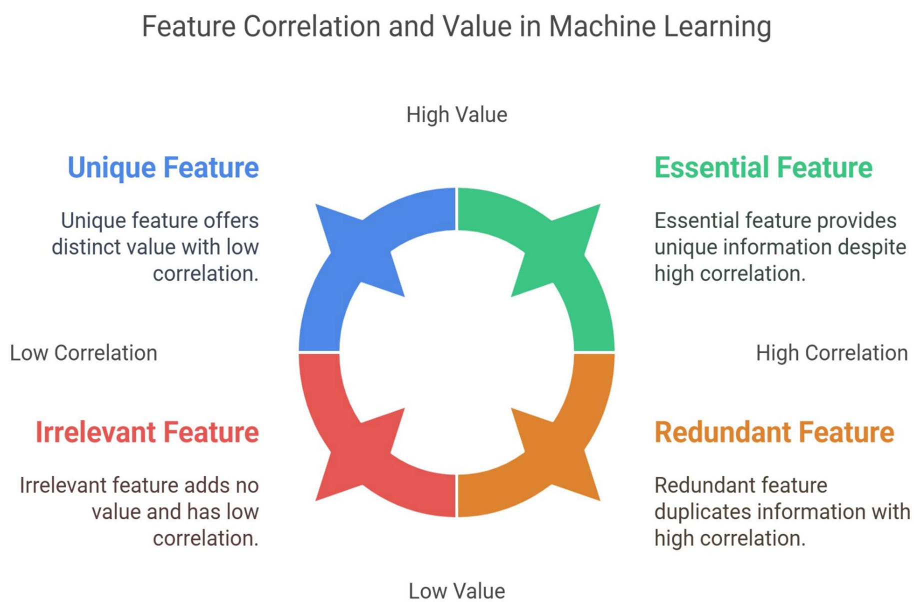
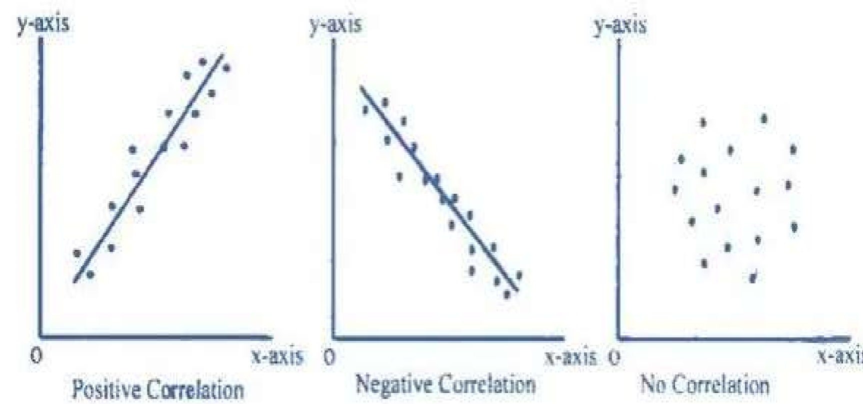
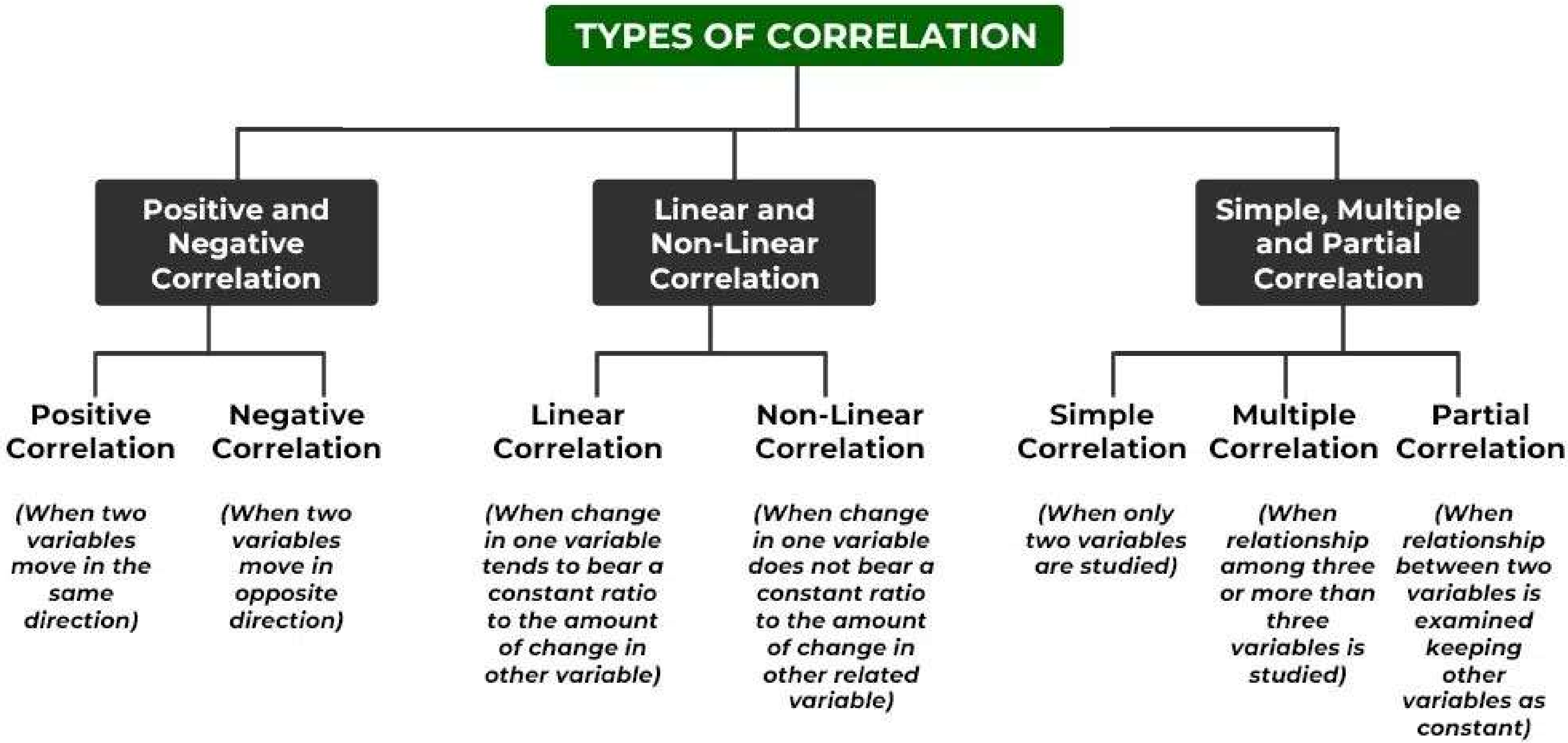
What is Correlation?

A statistical tool that helps in the study of the relationship between two variables is known as **Correlation.** It also helps in understanding the economic behaviour of the variables.The two variables are said to be correlated if a change in one causes a corresponding change in the other variable.

For example, A change in the price of a commodity leads to a change in the quantity demanded. An increase in employment levels increases the output. When income increases, consumption increases as well. The degree of correlation between various statistical series is the main subject of analysis in such circumstances.






TYPES OF CORRELATION
Positive and Negative Correlation
Linear and Non-Linear Correlation
Simple, Multiple and Partial Correlation
Positive Correlation
(When two variables move in the same direction)
Negative Correlation
(When two variables move in opposite direction)
Linear Correlation
(When change in one variable tends to bear a constant ratio to the amount of change in other variable)
Non-Linear Correlation
(When change in one variable does not bear a constant ratio to the amount of change in other related variable)
Simple Correlation
(When only two variables are studied)
Multiple Correlation
(When relationship among three or more than three variables is studied)
Partial Correlation
(When relationship between two variables is examined keeping other variables as constant)

1. Positive Correlation:

When two variables move in the same direction; i.e., when one increases the other also increases and vice-versa, then such a relation is called a **Positive Correlation.** ***For example,*** Relationship between the price and supply, income and expenditure, height and weight, etc.

Simultaneous increase in the value of both the Variables		Simultaneous decrease in the value of both the Variables	
X	Y	X	Y
20	120	60	500
30	170	50	400
40	220	40	300
50	270	30	200
60	320	20	100

2. Negative Correlation:

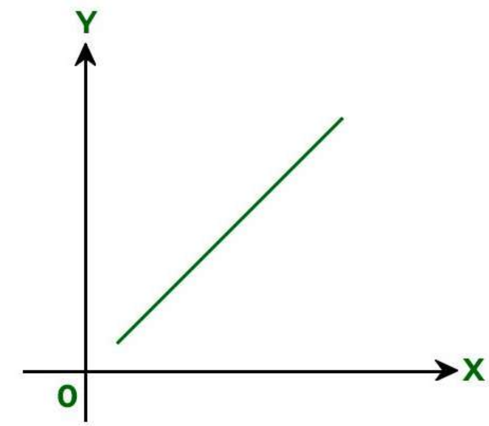
When two variables move in opposite directions; i.e., when one increases the other decreases, and vice-versa, then such a relation is called a **Negative Correlation.** ***For example,*** the relationship between the price and demand, temperature and sale of woollen garments, etc.

Rise in the value of one variable (X) and fall in other (Y)		Rise in the value of one variable (Y) and fall in other (X)	
X	Y	X	Y
10	100	250	100
20	90	200	k200
30	80	150	300
40	70	100	400
50	60	50	500

1. Linear Correlation: Based on the ratio of variations between the variables

When there is a constant change in the amount of one variable due to a change in another variable, it is known as **Linear Correlation.** This term is used when two variables change in the same ratio. If two variables that change in a fixed proportion are displayed on graph paper, a straight- line will be used to represent the relationship between them. As a result, it suggests a linear relationship.

X	10	15	20	25	30
Y	10	20	30	40	50

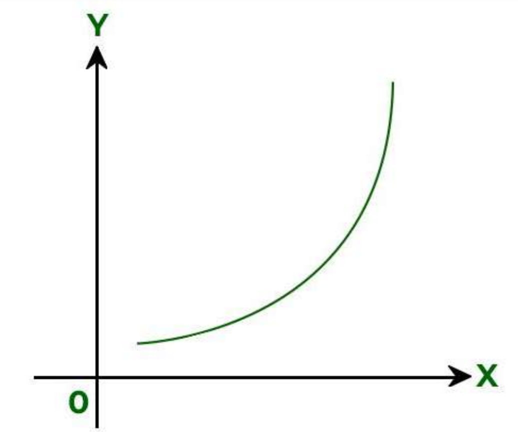


Linear Correlation

2. Non-Linear (Curvilinear) Correlation: Based on the ratio of variations between the variables

When there is no constant change in the amount of one variable due to a change in another variable, it is known as a **Non-Linear Correlation.** This term is used when two variables do not change in the same ratio. This shows that it does not form a straight-line relationship. ***For example*,** the production of grains would not necessarily increase even if the use of fertilizers is doubled.

X (Fertilizers)	10	20	30	40	50
Y (Production of Grains)	7	12	19	25	35



Non-Linear Correlation

Degree of Correlation

The degree of correlation is measured through the coefficient of correlation. The degree of correlation for the given variables can be expressed in the following ways:

Degree of Correlation

Degree of Correlation	Positive Correlation	Negative Correlation
Perfect Correlation	+1	-1
Very High Degree of Correlation	+0.9	-0.9
Fairly High Degree of Correlation	Between +0.75 and +0.9	Between -0.75 and -0.9
Moderate Degree of Correlation	Between +0.25 and +0.75	Between -0.25 and -0.75
Low Degree of Correlation	Between 0 and+0.25.	Between 0 and -0.25.
Zero/No Correlation (uncorrelated)	0	0

1. Perfect Correlation:

If the relationship between the two variables is in such a way that it varies in equal proportion (increase or decrease) it is said to be perfectly correlated. This can be of two types:

- **Positive Correlation:** When the proportional change in two variables is in the same direction, it is said to be positively correlated. In this case, the Coefficient of Correlation is shown as **+1**.
- **Negative Correlation:** When the proportional change in two variables is in the opposite direction, it is said to be negatively correlated. In this case, the Coefficient of Correlation is shown as **-1**.

2. Zero Correlation:

- If there is no relation between two series or variables, it is said to have zero or no correlation. It means that if one variable changes and it does not have any impact on the other variable, then there is a lack of correlation between them. In such cases, the Coefficient of Correlation will be **0.**

3. Limited Degree of Correlation:

There is a situation with a limited degree of correlation between perfect and absence of correlation. In real life, it was found that there is a limited degree of correlation.

- The coefficient of correlation, in this case, lies between +1 and -1.
- Correlation is limited negative when there are unequal changes in the opposite direction.
- Correlation is limited and positive when there are unequal changes in the same direction.
- The degree of correlation can be **low** (when the coefficient of correlation lies between 0 and 0.25), **moderate** (when the coefficient of correlation lies between 0.25 and 0.75), or **high** (when the coefficient of correlation lies between 0.75 and 1).

Pearson Correlation Coefficient

Pearson correlation coefficient is a measure of the linear relationship between two variables. It ranges from -1 to 1, where -1 indicates a perfect negative correlation, 0 indicates no correlation, and 1 indicates a perfect positive correlation.

Here's how to calculate the Pearson correlation coefficient in Python:

✅ **Formula**

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Where:

- x, y = variables
- n = number of observations

```python
import pandas as pd

# Load the data
data = pd.read_csv('data.csv')

# Calculate the Pearson correlation coefficient
corr = data['x'].corr(data['y'], method='pearson')

print('Pearson correlation coefficient:', corr)
```

Relationship between **Study Hours and Exam Marks**

Student	Study Hours (X)	Marks (Y)	X^2	Y^2	XY
A	2	50	4	2500	100
B	4	60	16	3600	240
C	6	70	36	4900	420
D	8	85	64	7225	680
E	10	95	100	9025	950

$$\sum X = 30, \quad \sum Y = 360, \quad \sum X^2 = 220, \quad \sum Y^2 = 27{,}250, \quad \sum XY = 2{,}390, \quad n = 5$$

$$r = \frac{5(2390) - (30)(360)}{\sqrt{[5(220) - 900][5(27250) - 129600]}}$$

$$r = \frac{11950 - 10800}{\sqrt{(1100 - 900)(136250 - 129600)}} = \frac{1150}{\sqrt{200 \times 6650}} = \frac{1150}{\sqrt{1{,}330{,}000}} \approx \frac{1150}{1153.26} \approx 0.997$$

✅ Interpretation

$$r \approx 0.997 \Rightarrow \text{Very strong positive correlation}$$

👉 More study hours → higher marks (almost linear).

Spearman’s Rank Correlation Coefficient

Spearman’s rank correlation coefficient is a measure of the monotonic relationship between two variables. It ranges from -1 to 1, where -1 indicates a perfect negative monotonic relationship, 0 indicates no monotonic relationship, and 1 indicates a perfect positive monotonic relationship.

Here’s how to calculate the Spearman’s rank correlation coefficient in Python:

✅ Formula

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Where:

- d = difference between ranks
- n = number of observations

```python
import pandas as pd

# Load the data
data = pd.read_csv('data.csv')

# Calculate the Spearman's rank correlation coefficient
corr = data['x'].corr(data['y'], method='spearman')

print('Spearman\'s rank correlation coefficient:', corr)
```

Used when **data is in ranks or ordinal** or not normally distributed

Marks of 5 students in two tests:

Student	**Test 1**	**Rank X**	**Test 2**	**Rank Y**	**d = X−Y**	**d^2**
A	90	1	85	2	-1	1
B	80	2	78	3	-1	1
C	70	3	72	4	-1	1
D	60	4	90	1	3	9
E	50	5	60	5	0	0

$$\sum d^2 = 1 + 1 + 1 + 9 + 0 = 12, \quad n = 5$$

$$\rho = 1 - \frac{6(12)}{5(25-1)} = 1 - \frac{72}{120} = 1 - 0.6 = 0.4$$

✅ **Result:**

$$\rho = 0.4 \Rightarrow \text{Moderate Positive Correlation}$$